



Neelam VidyaVihar, Sijoul, Mailam, Madhubani, Bihar – 847235

Website: <http://www.sandipuniversity.edu.in>

Email: info@sandipuniversity.edu.in

School of Computer Science & Engineering

Assignment: 02

Subject Name: Introduction to Machine Learning
Semester: IIInd

Subject code: AIML304
Branch: AIML

Short questions:

1. Define Machine Learning?
2. List any three real-life applications of Machine Learning?
3. What are the main processes involved in Machine Learning?
4. Differentiate between Supervised and Unsupervised Learning?
5. Give one example each for classification and regression problems?
6. What is Reinforcement Learning?
7. Name two algorithms used in Supervised Learning?
8. Name two algorithms used in Unsupervised Learning?
9. Define training dataset and test dataset?
10. What is the role of a model in Machine Learning?
11. State any two advantages of Machine Learning?
12. Mention one real-life example of Reinforcement Learning?
13. What is the purpose of feature selection in Machine Learning?
14. List two common challenges in implementing Machine Learning?
15. Give one example of Machine Learning in healthcare?

Long Questions:

1. Explain the various processes involved in Machine Learning with a neat diagram.
2. Discuss real-life applications of Machine Learning in different domains?
3. Compare Supervised, Unsupervised, and Reinforcement Learning with examples?
4. Describe the workflow of a Machine Learning project in detail?
5. Explain Supervised Learning. Discuss its types with suitable examples?
6. Explain Unsupervised Learning. Discuss clustering and dimensionality reduction techniques?
7. Explain the concept of Reinforcement Learning with a suitable real-life example?
8. Describe any three Machine Learning applications in business and industry?
9. Discuss the importance of data preprocessing in the Machine Learning process?
10. Explain the advantages, disadvantages, and limitations of Machine Learning?